HydroBoost – User Guide

1. Set the model input parameters in the Excel spreadsheets located in:

   Input_Spreadsheets/Model_A (or Model_B / Model_C)

2. In the *Simulation Configuration* tab of the input Excel spreadsheet, choose which cases to run and configure their parameters:

   - Set *Run_Flag* = *TRUE* for every row (case) you want to run.
   - Set the other columns (e.g., *Look_Ahead_Days, Interconnection Limits Inflow/Outflow, Max_Power, efficiencies*, etc.) per case.
   - The model reads and applies all BESS-related parameters (e.g., rated power, energy capacity, efficiencies) from this *Simulation Configuration* tab for each case.
   - During execution the model automatically iterates only over the cases with *Run_Flag* = *TRUE* and will save results in separate folders named after the *Case_ID*, e.g.:
     Simulation_Results/Model_C/Test 1/
     Simulation_Results/Model_C/Test 3/

   In the *Simulation Setting* tab, you can optionally set *test_num_days_value* to limit the simulation horizon for testing purposes:

   - If missing, blank, or 0: the model runs the full simulation year (365 or 366 days, depending on leap year).
   - If > 0: the model runs only the first N days (N = *test_num_days_value*).
   - If N exceeds the year length, it is automatically capped at 365/366.

3. Save the updated spreadsheet in:

   Input_Spreadsheets/Model_A (or Model_B / Model_C)

4. In *Core_Models/HydroBoost/src/HydroBoost_Sim.jl*, select the desired model by uncommenting the corresponding include line (lines 39-41):

   # include("model_structure/HydroBoost_Model_A.jl")
   # include("model_structure/HydroBoost_Model_B.jl")
   include("model_structure/HydroBoost_Model_C.jl") ← uncomment the one you want

5. Refresh market data forecasts. In a Python terminal, run:

   python Generate_forecast_price_data.py Model_C

   Replace Model_C with Model_A or Model_B as needed. This script automatically reads all required data for the selected model. No manual edits to any Python file are required.

6. In a Julia REPL, run the following (example with Model_C):

   import Pkg; Pkg.activate("."); Pkg.instantiate()
   include("Core_Models/HydroBoost/src/HydroBoost_Sim.jl")
   using .HydroBoost_Sim
   result = HydroBoost_Sim.execute_HydroBoost_model("Model_C")

   Replace "Model_C" with "Model_A" or "Model_B" as needed. This will automatically run all cases with *Run_Flag = TRUE* and write each case's results to its own folder under *Simulation_Results/Model_C/*.

7. Review the simulation outputs in the *Simulation_Results* folder.

Tip: use the same model name consistently across steps 4-6.

For any questions, doubts, or requests, please email: agrimaldi@anl.gov